实验三 顺摆系统辨识实验

一、实验目的

掌握系统辨识和参数估计的方法，设计利用实验建模获得系统模型和未知参数的方法。

二、实验原理

倒立摆时域模型

$\frac{J+ml^2}{ml}\ddot{\varphi}+\frac{c_r}{ml}\dot{\varphi}+g\varphi=-\ddot{x}$

频域模型

$G(s)=-\frac{mls^2}{(J+ml^2)s^2+c_rs+gml}$

辨识方法

$y(t)=-\ddot{x},\varphi_M(t)^T=[\ddot{\varphi},\dot{\varphi},\varphi],\theta=[\frac{J+ml^2}{ml},\frac{c_r}{ml},g]^T$

$\theta=(\varphi_M(t)^T\varphi_M(t))^{-1}\varphi_M(t)^Ty(t)$

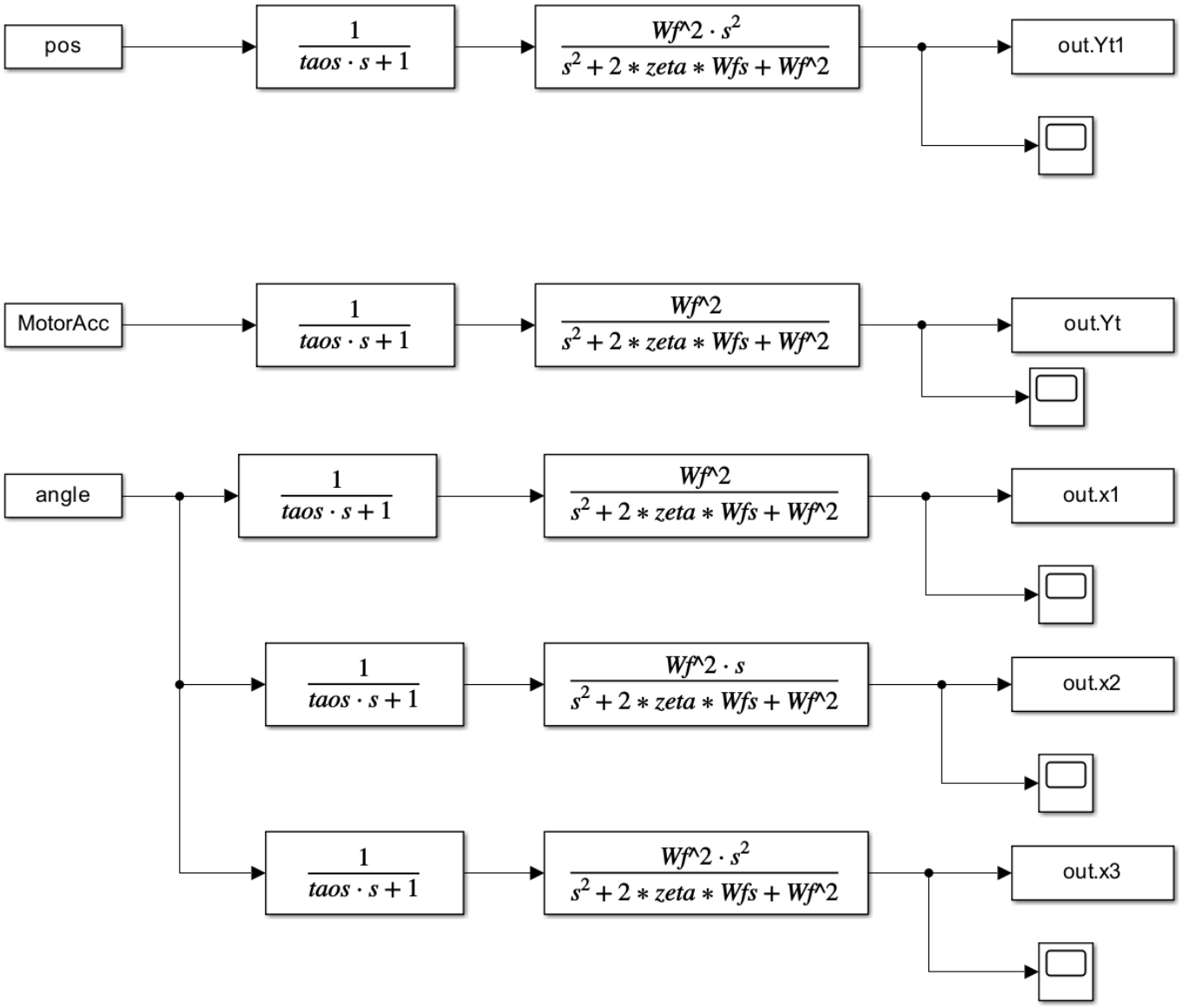
三、实验测试

系统辨识MATLAB源代码

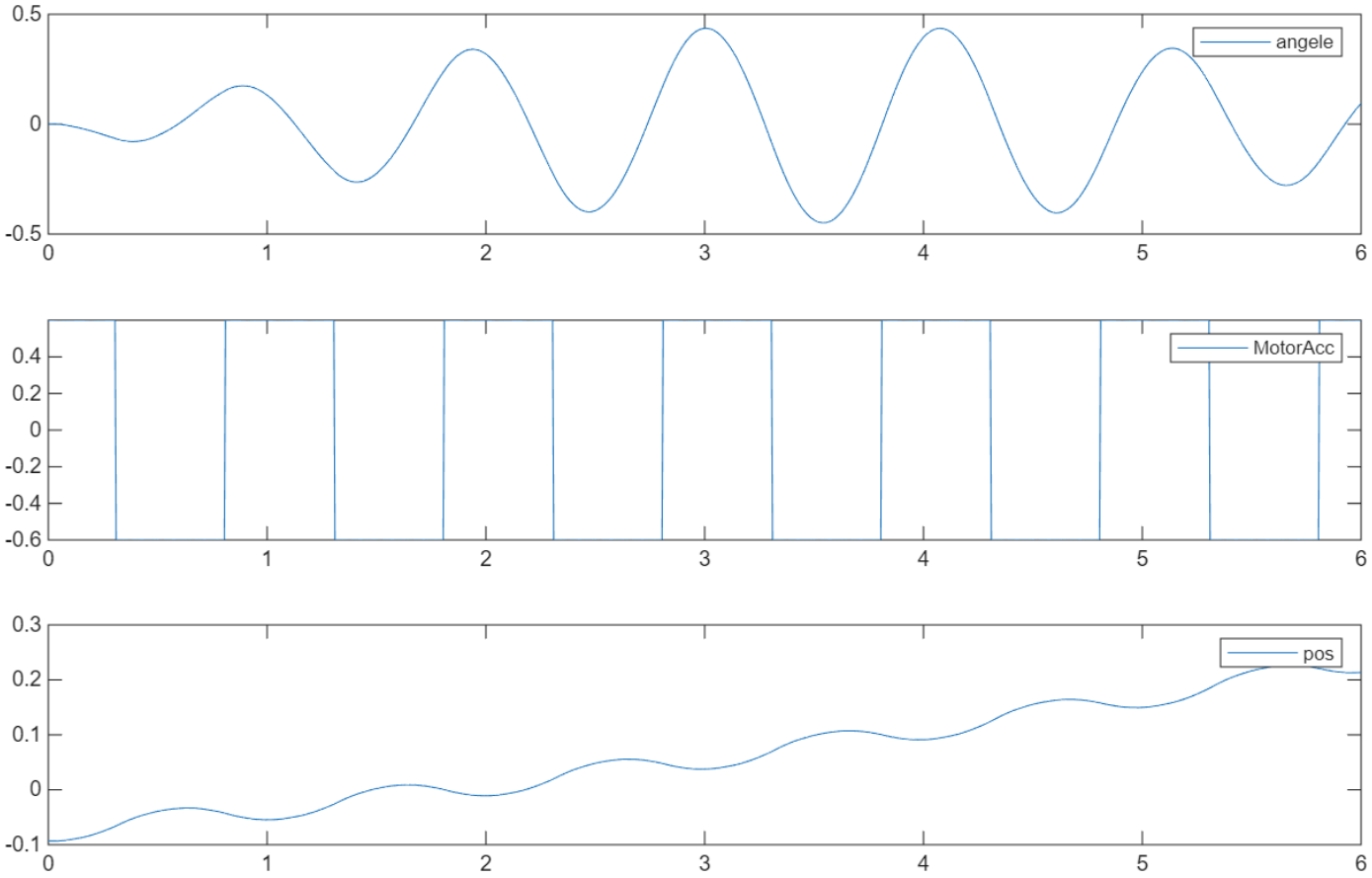
```
%% 辨识
clear;
clc;
close all;
```

```
%%
% data=load ('实测数据.txt');
taos=0.005;%角频率
zeta=0.9;%角阻尼
Wf=10;
dt=0.005;%采样时间
tdown=1;
tup=5;
time=6;

data=load ('实测数据.txt');
angle_data=(data(1:time/dt+1,1)+180)/180*pi;
pos_data=data(1:time/dt+1,2)/1000;
MotorAcc_data=data(1:time/dt+1,5);
len=size(angle_data,1);
t=0:dt:dt*(len-1);

MotorAcc=[t',MotorAcc_data];
angle=[t',angle_data];
pos=[t',pos_data];

%滤波
out=sim('Filter.slx');

%取出一段数据进行辨识
for k=tdown/dt+1:tup/dt+1
    MotorAcc_F(k-tdown/dt) = -out.Yt(k);%加速度滤波得到的加速度
    angle_F(k-tdown/dt) = out.x1(k);%滤波之后的角度
    angledot_F(k-tdown/dt) = out.x2(k);%滤波之后的角速度
    angleddot_F(k-tdown/dt) = out.x3(k);%滤波之后的角加速度
end
% 最小二乘法公式
phi_M = [angleddot_F' angledot_F' angle_F'];
theta = inv(phi_M'*phi_M)*phi_M'*MotorAcc_F';  %三个参数的辨识结果
Gs=tf(-1,[theta(1) theta(2) theta(3)]);
```

Simulink滤波文件



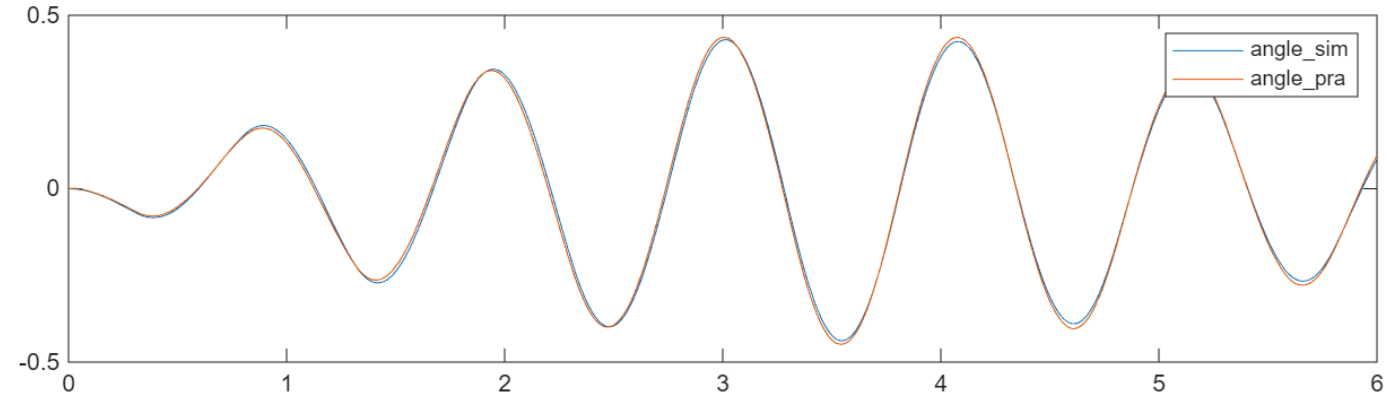
实测数据

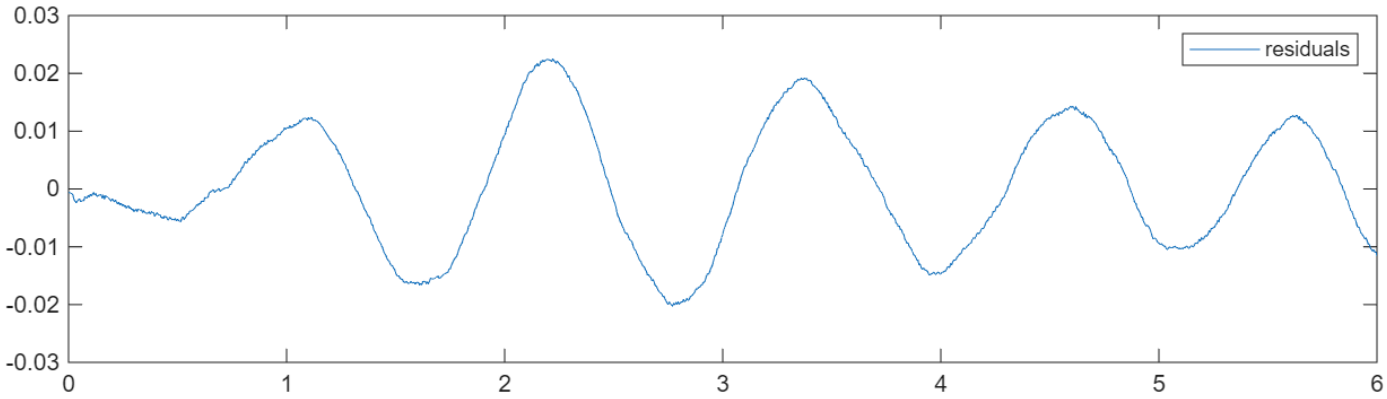


四、模型验证

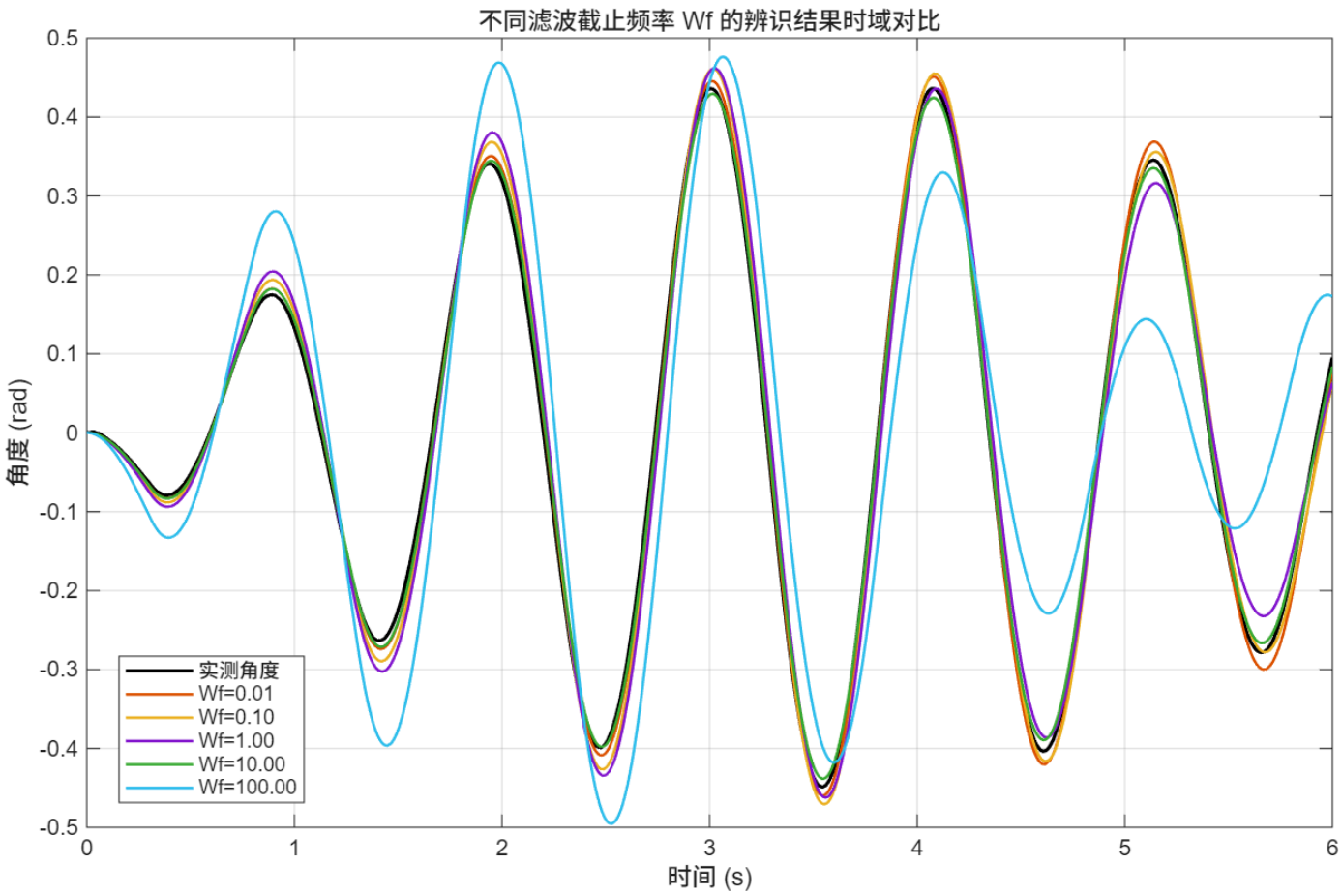
1. 时域模型辨识

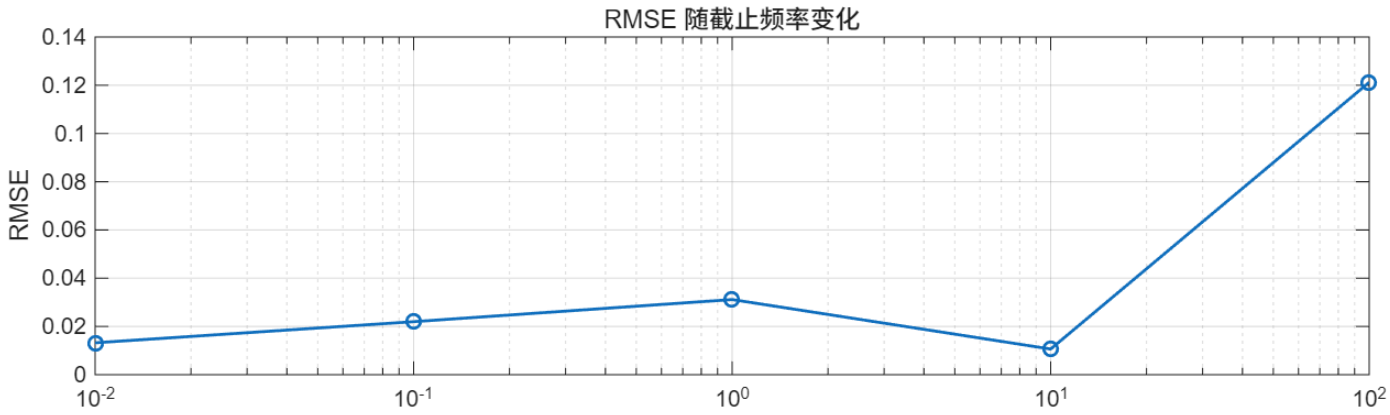
taos=0.005, zeta=0.9, Wf=10, tdown=1, tup=5时

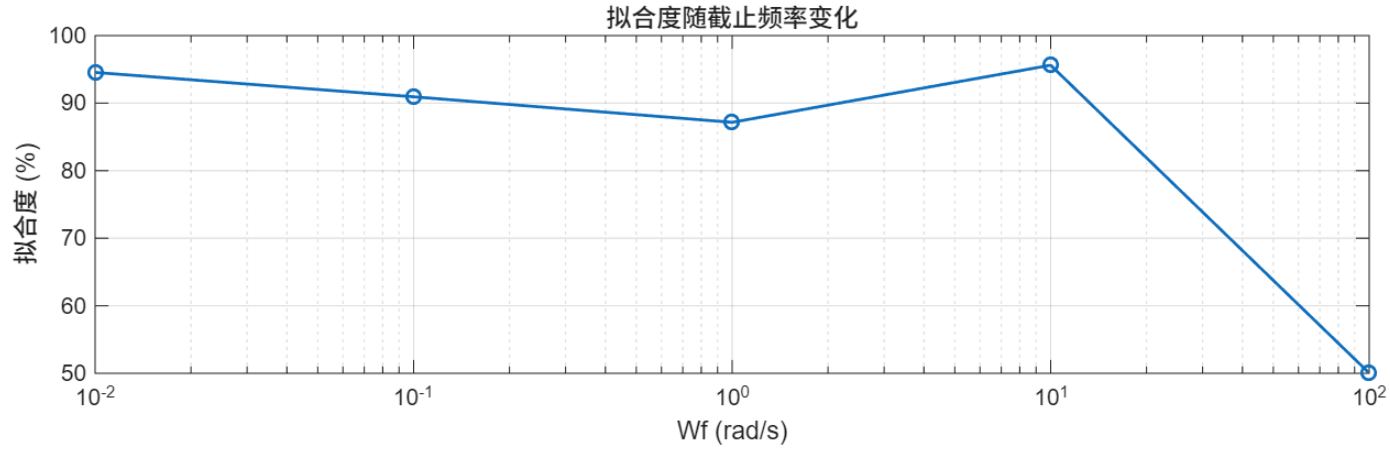




2. 保持输入激励相同，改变滤波截止频率

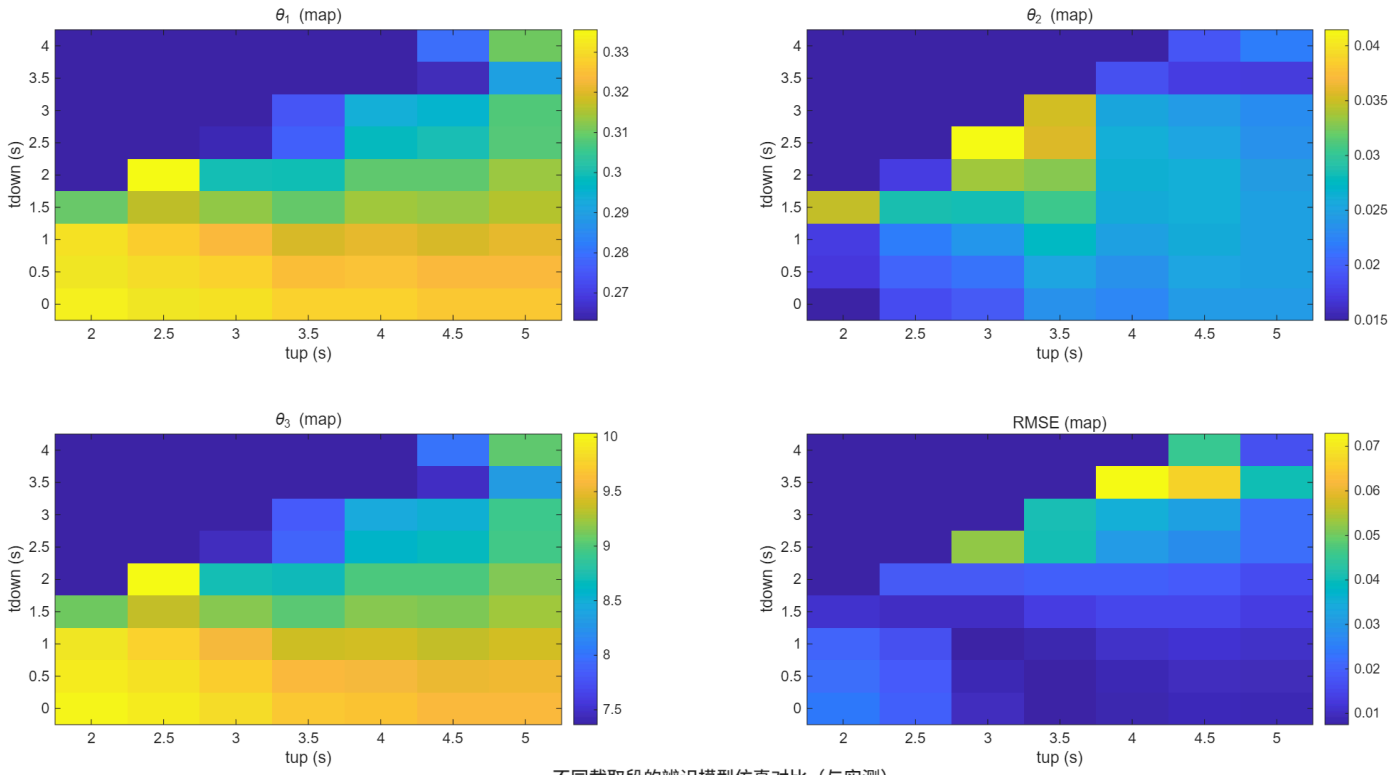


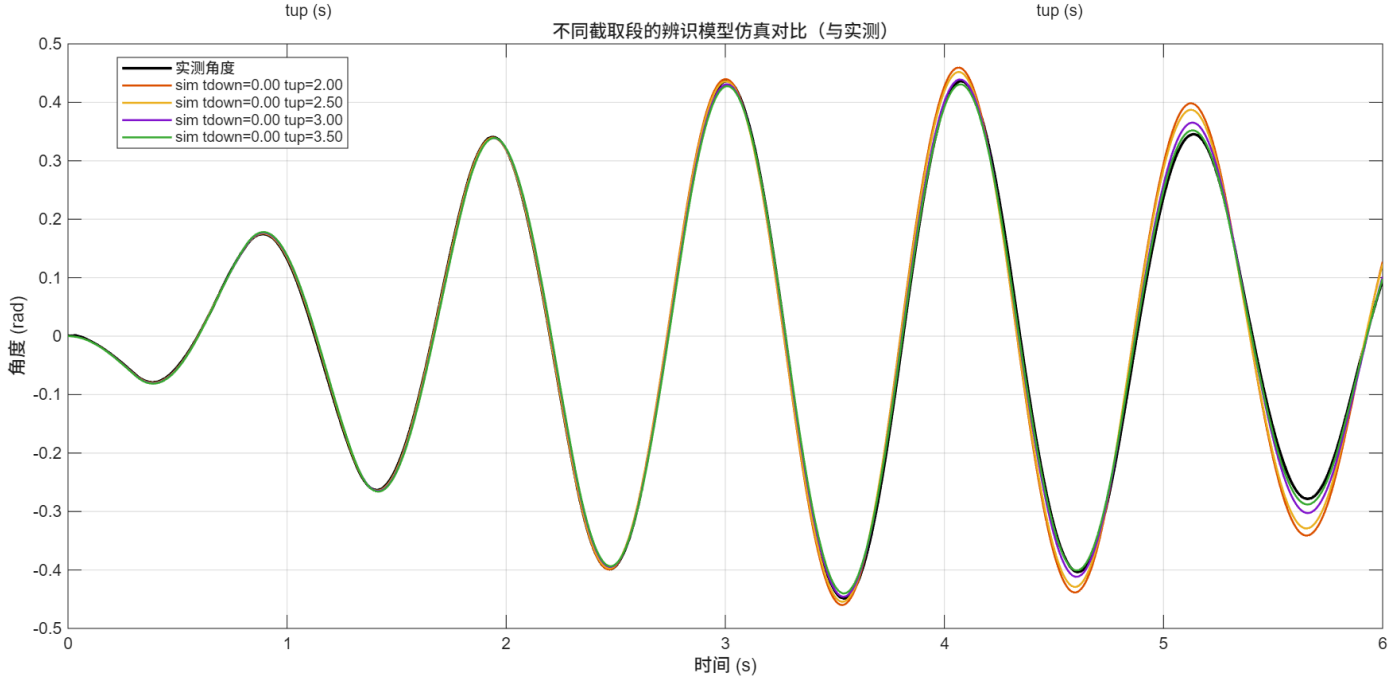




	θ_1	θ_2	θ_3
$W_f = 0.01$	0.3225	0.01234	9.524
$W_f = 0.10$	0.3042	0.01433	8.897
$W_f = 1.00$	0.2866	0.02496	8.264
$W_f = 10.0$	0.3206	0.02502	9.408
$W_f = 100$	0.2066	0.04327	5.467

3. 保持输入激励相同，改变数据截取





五、实验结果

1. 辨识结果

通过MATLAB进行时域模型的最小二乘估计，辨识结果与实测数据接近，残差基本在0.02rad（1.15°）以内。

2. 滤波截止频率的影响

从辨识结果对比图中看出，W_f过大时，辨识结果与实测数据差别较大；从辨识参数结果表中看出，θ_2即c_r有随W_f增大而增大的趋势，W_f过小时，θ_2很小，产生阻尼项的失真，但在辨识结果对比图中没有较大的影响。

3. 数据截取的影响

RMSE 最小值 0.00749，对应
$t_{down} = 1.00s, t_{up} = 3.00s, \theta = [0.323796, 0.023782, 9.57037]$

数据段过短时，方程条件数差、信息量不足，辨识结果不佳；包含开始阶段时，激励丰富但含有非线性、初始误差；包含中间阶段时，系统较为稳定，信噪较高，参数更准确。